

ILLUSTRATED INSTRUCTIONS

ED Throughput Sandbox Illustrated user guide

Instructions for the Sandbox and Visualizer workspaces.



Synchronized A/B maps in the Visualizer. Every patient shown is synthetic.

Compare scenario assumptions

Run a baseline, change one assumption, and compare outcomes across repeated simulations.

Open the app

alexkdot.github.io/ed-throughput-sandbox

Copy revised September 23, 2026. Instructions match the revised app source: Sandbox model v1 and Visualizer model v2.2.

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For education and hypothesis exploration. This model is not institution-calibrated and must not guide staffing, clinical, regulatory, or operational decisions.

START HERE

Your first run

Use Sandbox for aggregate results or Visualizer for patient and resource replay.

Sandbox	Visualizer
Compare demand, capacity and boarding using controls and charts.	Replay patient movement, inspect individual resources and compare two maps.
Use High / Moderate / Low acuity tiers. The reported window is 24 hours after a 24-hour warm-up.	Use ESI 1-5. The default replay is seven days after a 24-hour warm-up.
Reallocate fast-track spaces from the total treatment capacity.	Add fast-track spaces to main rooms and other care spaces.

Separate scenario models

Their assumptions, model definitions, saved setups and numerical results are separate. Switching workspaces does not convert a scenario. Compare A and B within the same workspace.

Run the Visualizer

- 1 Open Visualizer**
 Select **Open Visualizer** in the Sandbox header. In the Visualizer, **Sandbox** returns to the analytical workspace.
- 2 Start with A · Baseline**
 Use **Scenario settings** to review assumptions. Select **Run Scenario A** and wait for **Representative week ready**.
- 3 Replay, then inspect**
 Choose **Play** or **Jump to peak**. Pause and select a patient or resource. Scroll down to **Repeated-run outcomes**.

Run the Sandbox

In the Sandbox, start with **Scenario A** and the **Balanced baseline** preset. Select **Run Scenario A**. Read the median and 10th-90th percentile range, then follow pages 8-10 to compare an alternative.

A replication is one simulated run. A seed makes the random inputs repeatable. More replications produce more stable summaries of model variation; they do not establish real-world accuracy.

Set up the Visualizer

Use Show setup if Scenario settings is hidden. Choose the scenario to edit inside that panel.

Control	What changes
Average arrivals	Average synthetic arrivals per hour. Daily timing and ESI mix stay fixed unless changed elsewhere.
Main treatment rooms Fast-track spaces Hallway spaces	Rooms serve ESI 1-5; fast track serves ESI 4-5. Fast-track and hallway spaces are additional capacity.
Care-duration scale	Multiplies triage, treatment, diagnostics, reassessment and discharge durations. Inpatient delay is controlled separately.
CT scanners Lab processors	Changes parallel diagnostic capacity. A care space stays reserved while its patient is in imaging.
Off-room boarding spaces	Lets admitted patients release treatment spaces. Boarder census is uncapped; excess boarders keep holding care spaces.
Median inpatient delay	Changes post-admission delay. Inpatient units and bed matching are not simulated.

Diagnostic capacity check



Diagnostic capacity estimates for the default assumptions.

Balanced: below 85%.
Tight: 85% to below 100%.
Over capacity: 100% or higher.

These are approximate load estimates, available before a run. They flag potential capacity constraints before simulation.

Baseline configuration

The displayed baseline has 2 CT scanners and 4 lab processors. Review all assumptions, then run the selected view. Assumption inputs in Scenario settings are disabled during computation.

Adjust detailed assumptions

Open Advanced clinical-flow assumptions inside Scenario settings. The panel scrolls independently when its contents are long.

ESI mix, care time, and admissions

Changing one ESI share proportionally rebalances the other four to keep 100%.

Level	Mix		Initial care		Admit	
ESI 1	3	%	190	min	70	%
ESI 2	16	%	160	min	46	%
ESI 3	43	%	125	min	22	%
ESI 4	30	%	78	min	7	%
ESI 5	8	%	48	min	1.5	%

ESI mix, care time and admissions

Mix changes the share at each ESI level. Editing one share rebalances the other four so the total stays at 100%.

Initial care is the ESI-specific treatment median, in minutes. **Admit** is the probability of admission, expressed as a percentage.

The screenshot shows the five-level editor. Values are synthetic assumptions, not clinical benchmarks.

More detailed controls	How to use them
Pathway care-time multipliers	Scale initial care for minor injury, medical, abdominal or behavioral-health pathways.
Other stage medians	Set triage, reassessment, CT, MRI, X-ray, ultrasound, lab and discharge-lounge durations.
Diagnostic-order probabilities	Change pathway-specific probabilities for lab and imaging orders. Read each label; these are conditional probabilities.

Model detail and run setup

Open this disclosure for **Replications** (10-100; default 40), **Shared seed** and **Generate new shared seed**. Seed and replication changes apply to A and B. Rerun to update their results.

Total length of stay emerges from care, queues and boarding; it is not entered directly. Staffing and cost are omitted. Added resources assume matching staff and support.

Replay and inspect

The map is one selected representative run. Use it to examine the sequence of events.

Restart

-1 event

Play

+1 event

Speed

30x

Jump to peak

Add bookmark

Playback controls. The week slider moves to a selected minute.

To...	Use...
Control playback	Play / Pause ; choose Speed 1×, 5×, 30× or 120×.
Examine a transition	-1 event / +1 event to move to the previous or next event.
Navigate the week	Drag the week slider; use Restart to return to the beginning.
Find peak census or bookmark a time	Jump to peak finds peak census in the representative replay. Add bookmark places a clickable marker on the timeline.

ESI 3

P0399

Ambulance arrival

PHYSICAL LOCATION

Ultrasound

PATHWAY

abdominal

DOOR TO ROOM

46 min

TIME IN ED

6.9 hr

Concurrent statuses

Imaging in progress

Recent patient events

Tue 7:40 PM

Ultrasound 1: ultrasound started

Tue 7:40 PM

Moved to Ultrasound 1

Tue 4:37 PM

lab completed

Tue 3:36 PM

Lab processor 2: lab started

Tue 3:23 PM

initial treatment completed

Tue 2:28 PM

Room 6: initial treatment started

Tue 2:28 PM

Moved to Room 6

Tue 1:52 PM

Moved to the waiting room

Patient detail at a paused replay moment.

Select a patient

Click a patient dot or an entry under **Patients**. Use the filter to find an ID, ESI or status. Inspect physical location, pathway, door-to-room time, time in ED, concurrent statuses and recent patient events.

Select a resource

Use **Resources** or click a resource square. Read its state, queue and assigned patient. **Reserved** means its patient is elsewhere, but the space remains unavailable.

Patient dots use the ESI legend. Squares represent resources. The boarding area groups all boarders; check the inspector to distinguish off-room locations from patients holding care spaces.

Compare two scenarios

Change one assumption to isolate its effect within the model.

1 Copy the baseline

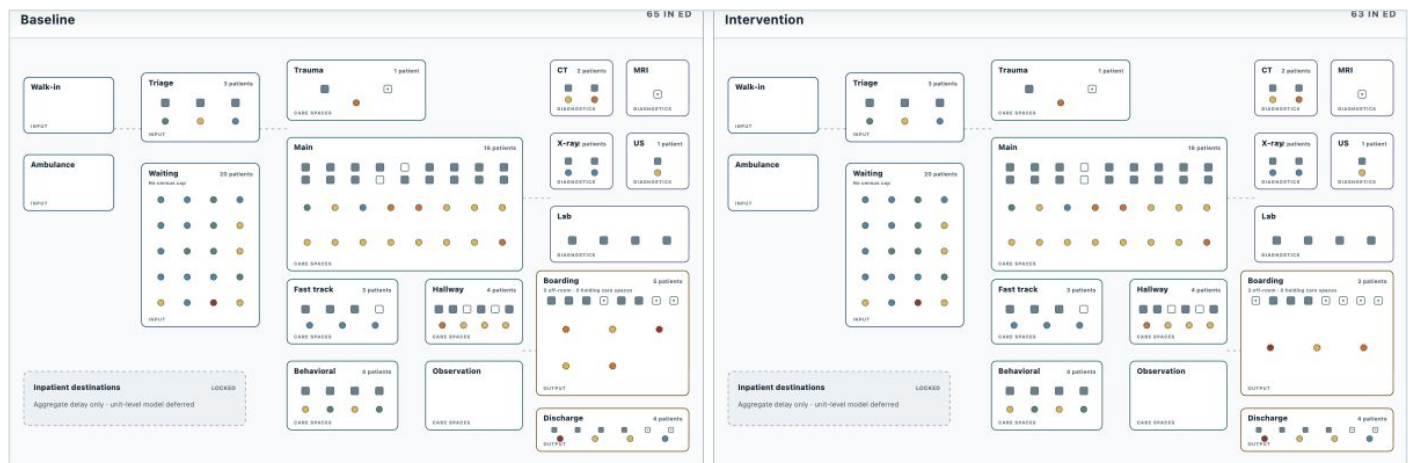
In Scenario settings, select **Copy A into intervention B**. This replaces B with A's assumptions and switches to B.

2 Change one assumption in B

Example: change **Median inpatient delay** from 260 to 180 minutes. Keep demand, seed and replications the same.

3 Run both together

Select **Side by side**, then **Run paired week**. Use **Hide setup** for more map space.



Worked example: A and B share a replay clock. The screenshots use seed 20260831 and 40 replications.

4 Inspect each scenario

Select a patient or resource in either map. Check **Inspect · Scenario A/B** so you know which scenario the detail panel describes.

5 Compare repeated-run outcomes

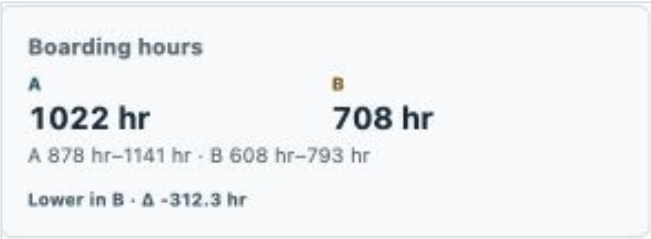
Scroll to **What changed**. Confirm the intended difference and pairing message, then read the outcome cards and uncertainty chart.

Demand pairing

With identical demand, the app pairs arrivals and synthetic patient attributes. If demand differs, it warns that pairing is by replication only; changes cannot be read as the isolated effect of capacity.

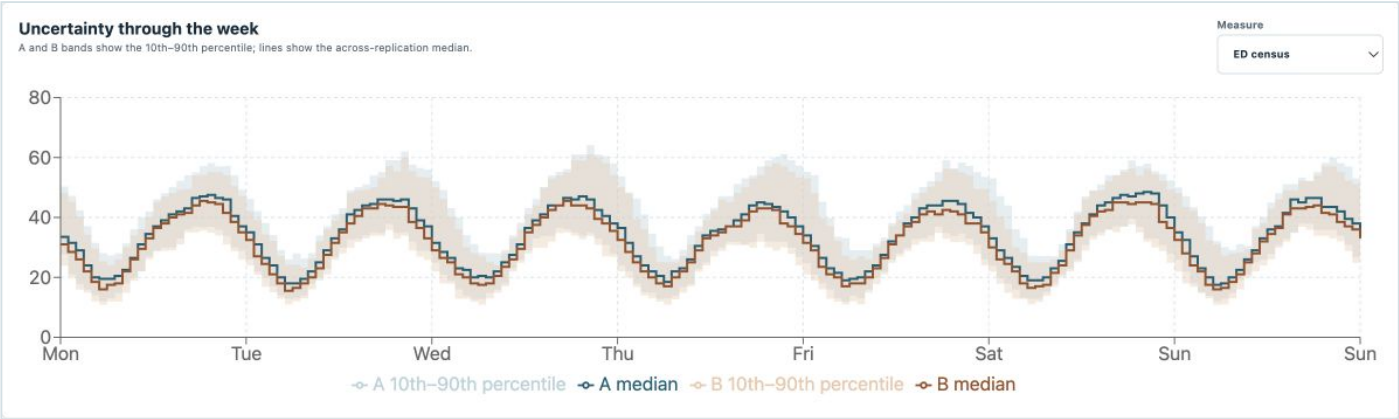
Interpret replay and aggregate results

Large values at the top follow the replay. Repeated-run outcomes summarize the full ensemble.



A and B: medians across replications.
Small ranges: 10th-90th percentiles.
 Δ : the median of paired B-minus-A differences.

For this example, boarding is lower in B. The paired change need not equal the difference between the two displayed medians.



Use Measure to choose ED census, Waiting, Boarders or Diagnostic queue. Lines show medians; shaded bands show 10th-90th percentiles.

Measure	Read it as...
Door to treatment space	Arrival to treatment-space assignment, summarized for treatment starts in the analysis window.
ED length of stay	Arrival to departure for patients departing in the analysis window.
Boarding / waiting patient-hours	Accumulated patient-time. Ten patients waiting for one hour contribute ten patient-hours.
Departures / peak counts	Departures in the window; highest ED census or waiting count within each replication.
Diagnostic queue delay	Median delay before diagnostic service. A zero median can coexist with some long waits.

The replay need not stay inside every ensemble band. Top-card time-series ranges use the nearest sampled time; the door-to-room range summarizes the full run. Ranges describe model randomness, not confidence in a hospital forecast.

Configure the Sandbox

The Sandbox uses a three-tier model and 24-hour analysis window.

SCENARIO A
Balanced ba...
NOT RUN

SCENARIO B
Intervention
NOT RUN

Balanced baseline

Duplicate A → B

Swap

Compare scenarios

More

▶ Run Scenario A

Choose A or B, name the active scenario, select comparison mode and run from this bar.

- 1 Start with a preset
- Choose **Balanced baseline**, Evening surge, Boarding bottleneck, Capacity constraint, Fast-track experiment or High-variability day. Presets are illustrative, not benchmarks.

Group	Inputs and meaning
Input	Set Mean arrivals , Arrival pattern and High / Moderate / Low acuity shares. Custom six-block profiles redistribute arrivals without changing the daily mean.
Throughput	Set Total treatment spaces and Treatment-time scale . Optional fast track reallocates spaces from the total for low-acuity patients; set its allocation and median duration.
Output	Set Weighted admission rate and Median boarding duration . Admitted patients keep occupying a treatment space while boarding.
Advanced assumptions	Set tier treatment medians, duration variability, tier admission probabilities, Monte Carlo replications (20-200) and Master seed . Default: 100 replications.

- 2 Run the chosen assumptions
- Select **Run Scenario A** or **Run Scenario B**. **Cancel run** stops computation without publishing partial results.

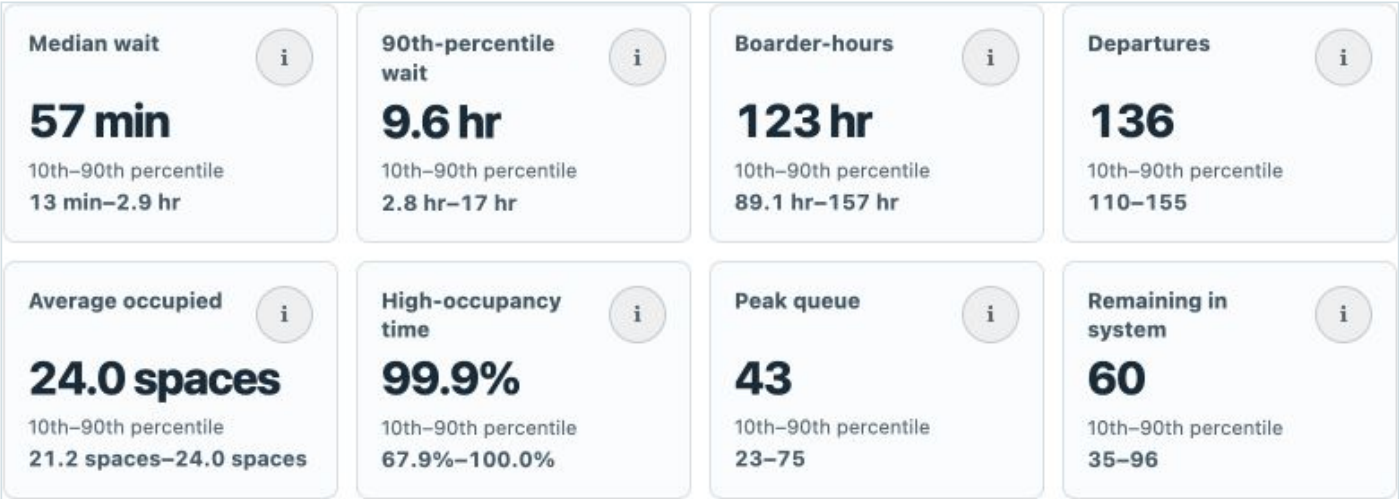
After you edit

The active tab can show **Changes not run**; results can say **Previous run**. Select **Run updated assumptions** before interpreting or exporting. Edits made during a run apply to the next run.

Seed and replication settings are shared by A and B. The same seed and assumptions reproduce the same results; New seed creates a different set of random inputs.

Read Sandbox results

Each main value is the median of a metric calculated separately in every replication.



Current baseline result cards. Select an i control to read the definition. Your results depend on the saved assumptions and seed.

Read together	Why it matters
Median wait and 90th-percentile wait	Arrival to an eligible space. The 90th-percentile wait is calculated within each run; the range beneath it describes variation across runs.
Boarder-hours and Average occupied	Boarding consumes treatment capacity. Occupied spaces include active treatment and boarding.
High-occupancy time	Share of analysis time at or above 90% of configured capacity.
Departures, Peak queue and Remaining in system	Check completed work alongside the largest queue and unfinished workload. A wait metric alone misses people still waiting.

Charts and acuity results

System status over time shows occupancy and waiting in 15-minute time-weighted bins. Open **Read chart data** for the table. **Arrival and departure pattern** shows hourly event counts; open **Read hourly flow data**. **Wait by acuity** separates High, Moderate and Low.

Measurement cohorts

Warm-up patients may start treatment or depart during analysis; new arrivals can remain unfinished at its end. Waits, arrivals, departures and length of stay do not describe one identical patient cohort. **N/A** means no eligible observation.

Compare scenarios and run sensitivity

Use a paired comparison for an alternative and a sensitivity sweep to examine one input over several values.

1 Duplicate A into B

Select **Duplicate A** → **B** and confirm replacement if prompted. Change one assumption in the new **Intervention** scenario.

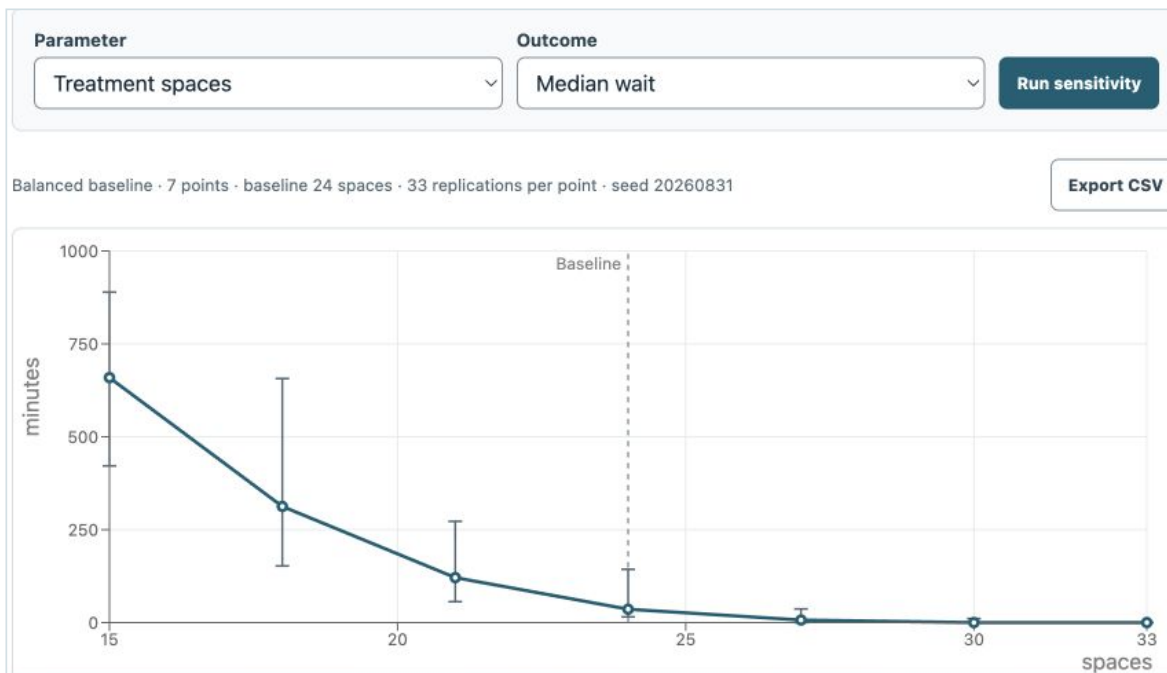
2 Compare A + B

Enable **Compare scenarios**; the control reads **Comparing A + B**. Select **Run both scenarios**. Read **Changed assumptions** and **Paired change**.

Paired change is **B minus A**, computed within each aligned replication before aggregation. Negative means lower in B. **Swap** exchanges the scenarios and reverses the comparison direction.

Run a sensitivity sweep

Select **Sensitivity** in the header. Choose a **Parameter** and **Outcome**, then select **Run sensitivity**. Review the baseline marker, median line, range bars and data table. Use **Export CSV** to save the sweep.



Example: treatment spaces versus median wait. The default 100-replication setting uses 33 replications per sweep point.

Sensitivity scope

Runs up to seven valid values using 20-40 replications per point. Each sweep changes one parameter and does not test interactions. Rerun if the scenario, parameter or outcome changes.

SAVE AND EXPORT

Save, share and export

Save assumptions for reproducibility and export completed results for review.

Action	Sandbox	Visualizer
Save assumptions	More → Export scenario JSON ed-throughput-scenarios.json	Export setup ed-visualizer-scenarios-v2.json
Restore assumptions	More → Import scenario JSON Up to 64 KB; schema v1.	Import setup Up to 256 KB; schema v2.
Export results	Export CSV in the results heading. A/B results are included when comparing; the paired-delta table is not.	Export comparison report (.csv) in Repeated-run outcomes. Includes ensemble metrics, hourly ranges, paired deltas and provenance.
Share / print	More → Copy share link shares assumptions, not results. Print report opens browser printing; choose Save as PDF.	Use setup JSON to share assumptions and CSV to share outcomes. The Visualizer has no dedicated share-link or Print report control.



Import replaces the current setup

Choose the correct workspace and file. Review the replacement prompt before confirming: both scenarios are replaced and completed results or replay are cleared. Rejected files leave the current setup in place.

The two JSON formats are not interchangeable. Export a backup before importing another setup.

Data retained after reload

Assumptions are saved in browser storage where available. Completed results, replay and bookmarks are not saved across reloads; run again. Bookmarks also clear with a new run or import. Use JSON files as durable backups. Keep patient information and confidential details out of scenario names and inputs.

Sandbox share links are snapshots. If later edits revert on reload, return to the base app URL or remove the `?state=` query, then create a new link. If copying fails, copy the link placed in the address bar.

REFERENCE

Troubleshooting and reference

Use the table below to resolve stale results, disabled controls and import errors.

What you see	What to do
Changes not run / Previous run	Rerun after editing. In the Visualizer, stale results disable export and peak shortcuts.
Replay controls are disabled	Run the selected scenario first. Choose Side by side and Run paired week to populate both maps.
N/A or zero	N/A means no eligible observation. Zero may be a valid or rounded value. At the start of a replay, door-to-room can be N/A until a new treatment start occurs.
Lower in B · Δ 0 min	A small negative change can round to zero minutes. The direction uses greater precision; inspect the CSV for numerical detail.
Import rejected	Check the workspace, schema, file size and allowed values. Restore a file exported from that workspace.
Run fails or is cancelled	Run again. If failure repeats, reload and use fewer replications or less extreme assumptions. No partial result is published.
Queues or waits become extreme	Review arrivals, care duration, diagnostic capacity and boarding together. Strict priority can produce very long lower-acuity waits under overload.
Controls do not fit on screen	Use the Sandbox domain buttons and bottom run controls. The Visualizer is designed primarily for desktop use.

Keyboard and display

Use **Tab** / **Shift+Tab** to move through controls and **Enter** / **Space** to activate buttons. In the Sandbox, arrow keys navigate scenario tabs and **Escape** closes menus and dialogs. Use chart data tables for text access. In the Visualizer, choose **Motion** → **Reduced** if helpful.

Version and source notes

Copy and instructions revised September 23, 2026. Updated labels were checked in the local app. Simulation behavior remains edts-model-v2.2 | aggregate-v2.2.

For definitions and limits, open **Methodology** in the Sandbox or read the [Sandbox model](#) and [Visualizer model](#) documentation. Screenshots are examples, not performance targets. Staffing, cost and institution-specific calibration are outside the model.